Zihan Yuan

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EDUCATION

Rice University — *Houston, Texas*
Master of Computer Science — *Jan 2026 – Dec 2027*

Southern University of Science and Technology — *Shenzhen, China*
Bachelor of Science in Computer Science | GPA: 3.73/4.00 (90%) — *Aug 2021 – Jul 2025*

WORK EXPERIENCE

Parse Health (parse-health.org) — ***Full-Stack Engineer***

Intelligent Medical Research Assistant (Ontology & WHI Agents) — *Jul 2025 – Present*

- Developed a **LangGraph**-based **Ontology Agent** with **RAG** to **generate SQL queries** and help medical students navigate complex clinical terminology.
- Built the **React research workspace** for the **UMLS Agent**, with **SSE streaming chat**, **bilingual structured replies** for SQL, concept clarification, hierarchy, and sourced knowledge, and **per-turn run traces** that remain inspectable after refresh.
- Added **CSV cohort upload** with column filters, distributions, **batch CUI lookup**, and **hierarchy-union tables**, plus **shareable read-only sessions** with file preview and download, **voice input**, and a responsive registration flow that collects a **UMLS API key**.
- Built the **WHI Agent** for **cross-cohort harmonization** of WHI and MESA (dbGaP), using biomedical encoders (**PubMedBERT** and **SapBERT**) to retrieve variables and a **covariate-adjusted ranker** that blends semantic similarity with distribution-tail distance across **14 study accessions**.

KAGE — Hematology Knowledge Graph Explorer — *Aug 2025 – Present*

- Took over the existing **KAGE** explorer and replaced its WebGL view (**3d-force-graph** / **Three.js**) with a custom **Canvas 2D** renderer. Projected a **d3-force-3d** layout through a perspective camera, clustering same-category concepts into islands so **PheCode**, **RxNorm**, **LOINC**, and **UMLS** networks stay readable under orbit and hover.
- Adapted the **Vue 3** frontend to the new **KAGE API** backed by **Spring Boot**, fixing **search, legend, and layout bugs**.

MSTech (mstech.ai) — ***Technical Partner & CTO***

Strategic partnership with Posit Software, PBC (formerly RStudio) — *Nov 2025 – Present*

- Led the **technical validation and deployment strategy** for enterprise analytics solutions, enabling the firm to become the **first Professional Services Partner (PSP) for Posit in the Asia-Pacific region** and strengthening its regional market leadership.
- Played a key role in the **Posit Bootcamp** and passed the **Posit Partnership Technical Engineer Assessment** with a **high score**, helping the firm become **China's only Posit Premier Partner**.
- Directed the localized **Kubernetes** deployment and **private network debugging** of data science clusters for **Qilu Pharmaceutical**, overcoming **domestic cloud infrastructure constraints** and providing technical support for an engagement valued at over **CNY 3.5 million**.
- Built an **OpenTelemetry (OTEL)** pipeline to monitor Qilu Pharmaceutical's service health: **OTEL Collector** for telemetry collection, **Prometheus** for metrics, custom **Grafana dashboards**, **Loki** for log collection, and **Jaeger** for OTEL **distributed tracing**.
- Built an **operations ticketing platform** and an **education platform** for **Qilu Pharmaceutical** to support service requests and user training.

Unified Enterprise IT & DevOps Ecosystem — *Aug 2025 – Jan 2026*

- Deployed **Dockerized Keycloak** with **OIDC** across **nine internal platforms**, standardizing authentication and reducing employee onboarding time by **87%** while strengthening security governance.
- Built **CI/CD pipelines** with **Coolify** and self-hosted **GitLab Runners** to **automate application builds and deployments**.
- Extended open-source **Kimai** with **twelve custom plugins** for timesheets, leave and holiday approvals, SOP document control, personnel compliance, and task review behind **Keycloak single sign-on**, replacing spreadsheet-and-email workflows with a unified system of record for professional-services delivery.
- Embedded **LLM summarization** on live timesheet data to generate **daily, weekly, and project activity reports** with **scheduled delivery**, giving leadership a continuous view of utilization and client-delivery progress without manual status collection.

PROJECTS

AI Doctor — Digital Intelligent Physician | AI Agent & Full-Stack Engineer — *Aug 2025 – Oct 2025*

- Built an interactive **digital physician app** with an **AI orchestration backend** for structured consultations and **agent-driven diagnostic guidance**.
- Designed **multi-agent RAG** for **multi-step clinical reasoning**, **symptom triage**, and **context-aware medical Q&A** across longitudinal patient inputs.

AquaGuard — Visual Analytics System for VAST Challenge 2024 | Primary Author & Front-end Engineer — *Nov 2023 – Jun 2024*

- Designed and implemented **AquaGuard** for **VAST Challenge 2024**, enabling investigation of **illegal fishing** in large maritime datasets through **vessel filtering, side-by-side comparison, anomaly review, and drill-down exploration**.
- Built a multi-view frontend with **D3.js, Vue 3, and ECharts**, linking **knowledge graphs, temporal diagrams, heatmaps, vessel trajectories, and hierarchical trees**.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, TypeScript, JavaScript, SQL

AI & Data: LangGraph, RAG, Multi-Agent Systems, Biomedical Encoders (PubMedBERT, SapBERT), Data Harmonization

Backend: Flask, Spring Boot, Node.js, REST APIs, OAuth2, JWT

Frontend: React, Vue 3, HTML/CSS

Databases: MySQL, PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps: AWS, Kubernetes, Docker, Linux, Git, GitLab CI/CD, Coolify, Nginx, Caddy, OTEL, Prometheus, Grafana, Loki, Jaeger